

Do LLMs Give Consistent Opinions? Evaluating Response Reliability Under Varying Likert-Scale Formulations in Survey-Style MCQA

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Abstract

Survey methodology requires that equivalent question formulations elicit stable responses from the same respondent, regardless of how the Likert-type answer scale is presented. As LLMs are increasingly deployed as synthetic survey respondents, a critical but underexplored question arises: do they satisfy this basic *parallel-forms reliability* standard when answer scales vary in wording, cardinality, polarity, or option order? We extend the OpinionQA into two stress-test benchmarks, 1,235 subjective questions each paired with five reworded scale variants and six positional permutations, evaluated across four prompt-label formats. Using a four-metric consistency framework, we benchmark eight open-weight 7B-9B models from four families in both base and instruction-tuned configurations. Three findings emerge. First, base models fail basic reliability requirements: they exhibit severe mechanical instability driven by primacy bias, label sensitivity, and formatting failures, including full polarity inversions for the same question. Second, instruction tuning acts as a reliability filter, reducing response variance by approximately 50% and enabling models to treat semantically equivalent scales as equivalent. Third, for instruction-tuned models under randomized option ordering, removing explicit labels paradoxically *increases* consistency by eliminating symbol-binding conflicts. These findings establish that response reliability is an engineered property of LLMs requiring both alignment training and deliberate prompt design, with concrete implications for survey-like MCQA evaluations. ¹

1 Introduction

The reliability of survey measurement instruments is a cornerstone of empirical social science. Classical psychometric theory formalizes this through

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¹Data and code are available at: <https://github.com/boleima/SurveyScaleQA>.

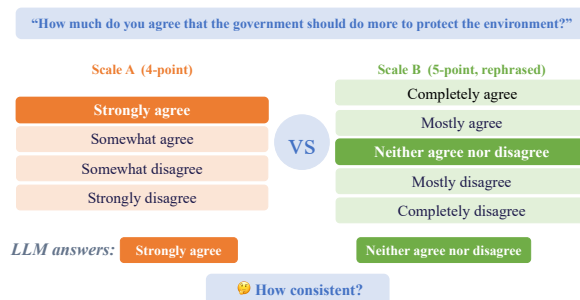


Figure 1: Example of LLMs in responding to a question with different scales. Are they consistent?

the concept of *parallel-forms reliability*: equivalent versions of a survey question, differing only in surface-level scale properties, should produce comparable responses from the same respondent (Lord et al., 1968; Cronbach, 1951). As LLMs are increasingly deployed as synthetic survey respondents (Aher et al., 2023; Santurkar et al., 2023), this standard becomes directly testable. Because subjective survey items lack a ground-truth “correct” answer, standard measures of *validity* (or accuracy) are undefined. Therefore, this paper strictly evaluates *reliability* (self-consistency). A reliable LLM respondent should express the same underlying opinion regardless of whether the scale uses four or five points, whether agreement is worded as “Strongly” or “Fully”, or whether the positive end of the scale appears first or last. Figure 1 shows an example of an LLM responding to the same question under different scale configurations.

Evidence on LLM format sensitivity suggests this requirement is far from guaranteed. Models exhibit systematic positional biases (Zheng et al., 2024; Wang et al., 2024a), fail to robustly bind answer content to option labels (Xue et al., 2024), and produce volatile outputs under minor prompt perturbations (Ngweta et al., 2025; Lunardi et al., 2025). Several studies document that LLM survey responses shift substantially with scale redesign

(Rupprecht et al., 2025; Khan et al., 2025; Lee et al., 2025). However, these studies typically examine a single model family, do not manipulate scale design as systematically as survey researchers would, and rarely isolate the role of alignment training as a driver of reliability.

This paper addresses these gaps through a controlled, multi-factor evaluation grounded in survey methodology. We vary three dimensions of Likert-scale design: (1) semantic wording and cardinality, (2) option order, and (3) prompt-label format. We evaluate all combinations across eight models from four families, comparing Base and Instruction-Tuned variants. Crucially, to strictly isolate the effects of prompt format and scale design from stochastic generation noise, we enforce greedy decoding across all experiments; our measurement of “consistency” therefore captures deterministic structural sensitivity rather than sampling variance.

Contributions. This paper makes the following contributions:

- We release an extended OpinionQA benchmark with systematic Likert-scale variations (6,175 Reworded and 7,410 Shuffled question-scale pairs, each across four prompt formats), designed as a reusable consistency evaluation resource.
- We introduce a four-metric consistency framework that disentangles average drift (APD, MSD), worst-case divergence (MPD), and cross-format structural stability (Pearson r), grounded in the psychometric concept of parallel-forms reliability.
- We provide the first large-scale comparison of Base vs. Instruction-Tuned models across Likert scale design dimensions, establishing that for models in the 7B-9B parameter class, alignment is the primary reliability determinant and reduces response variance by $\approx 50\%$.
- We show a counterintuitive pattern: when option order is randomized, removing labels actually makes instruction-tuned models *more* consistent, not less, because it avoids a conflict between a model’s habitual label choice and the option it actually means to pick, an interaction between prompt format and option order that prior work has largely overlooked.

2 Background and Related Work

Survey reliability and Likert-scale design. Classical Test Theory decomposes an observed sur-

vey response into a true latent score and a measurement error component (Lord et al., 1968). A key source of measurement error in Likert-scale surveys is scale design itself: the number of response options, the directionality (unipolar vs. bipolar), and verbal labels attached to scale points have all been shown to shift human responses independently of the underlying attitude (Menold, 2021; Höhne et al., 2022). Human respondents also exhibit well-documented response styles, including acquiescence bias (Couch and Keniston, 1960), satisficing (Krosnick, 1991), and order effects (Krosnick and Alwin, 1987). Parallel-forms reliability addresses this challenge by requiring that two equivalent instrument versions, differing only in surface properties, yield correlated results (Cronbach, 1951). This is precisely the standard we apply to LLMs: a model queried with semantically equivalent scales should express the consistent underlying opinion.

LLM format sensitivity and symbol binding.

Accumulating evidence shows that LLM responses in multiple-choice settings are highly sensitive to presentation format. Models exhibit primacy and recency biases, with GPT-3.5’s accuracy on MMLU dropping by over 6 percentage points when the correct answer is consistently placed last (Zheng et al., 2024). First-token probabilities and text-output responses frequently disagree for the same prompt, with mismatch rates exceeding 60% for instruction-tuned models (Wang et al., 2024b), and minor formatting changes can cause accuracy fluctuations exceeding 76% (Ngweta et al., 2025). Xue et al. (2024) trace these failures to inadequate *symbol binding*, namely the inability to reliably associate option content with its assigned label, a failure mode that persists through fine-tuning. Molfese et al. (2025) further show that evaluation procedures can confound reasoning failures with formatting failures, underscoring the need to evaluate response consistency separately from accuracy.

LLMs as synthetic survey respondents. When LLMs act as virtual survey respondents, format sensitivity becomes a validity threat. Rupprecht et al. (2025) systematically perturb World Values Survey items and find consistent human-like biases, including strong recency effects, in LLM responses. Khan et al. (2025) show that apparent cultural alignment signals are largely methodological artifacts of prompt variation rather than stable model traits. Tjuatja et al. (2024) find that LLMs generally fail

to replicate human response biases in controlled survey settings, and Lee et al. (2025) demonstrate that even high-performing models are not consistent evaluators across rating scales. Santurkar et al. (2023) found that base models tend toward liberal-leaning opinions and are difficult to steer toward specific demographic views, a finding whose implications for opinion stability this paper directly addresses.

3 Methodology

3.1 Dataset Construction: How We Vary the Scales

Source Data. We build on the OpinionQA corpus (Santurkar et al., 2023), derived from the Pew American Trends Panel (Pew Research Center, 2024). After manually filtering questions requiring specific respondent personas or responses that cannot be mapped to Likert-type scales, we retain $N = 1,235$ highly subjective questions. Each question is classified into one of 15 semantic *content dimensions* (e.g., Agreement, Importance, Frequency, Better/Worse), which determine which family of Likert scales is appropriate for generating variations. The full distribution is in Appendix A.

Reworded Dataset: same question, different scale. For every question, we write five answer scales that all ask the same thing but differ along three dimensions that survey designers routinely choose between: how many response points the scale has (**cardinality**: 4, 5, or 6 points), which words label those points (**phrasing**, e.g., “Strongly” vs. “Fully” vs. “Completely”), and whether the scale runs from negative to positive or from zero to high (**polarity**: bipolar vs. unipolar). Table 1 shows this concretely for one question: V1 and V2 are both 4-point bipolar scales that only swap the wording (“Strongly/Somewhat” vs. “Fully/Partially”), while V3 keeps similar wording but moves to five points and adds a neutral middle option. Repeating this across all 1,235 questions gives a Reworded Dataset of 6,175 question-scale pairs ($1,235 \times 5$), where the question itself never changes and only the scale wrapped around it does.

Shuffled Dataset: same scale, different order. To isolate the effect of option order on its own, we take one representative scale per question and generate five random permutations of its option order, using fixed seeds (2, 7, 10, 67, 101), and then add the original ordering back as a sixth version.

The bottom row of Table 1 (S1) shows one such shuffle of V1: the wording and number of options are identical to V1, but “Strongly agree” now sits in the third position instead of the first. This gives a Shuffled Dataset of 7,410 pairs ($1,235 \times 6$), where wording and scale length are held fixed and only the position of each option varies.

Response Normalization. Because we need to compare answers across scales with different numbers of points, we map every valid response onto the same $[-1, 1]$ interval, where $+1.0$ marks the most positive option (e.g., “Strongly Agree”) and -1.0 the most negative. Options are spaced evenly along this interval: a 4-point scale maps to $\{-1.0, -0.33, 0.33, 1.0\}$, a 5-point scale to $\{-1.0, -0.5, 0.0, 0.5, 1.0\}$, and so on, as shown in the rightmost column of Table 1. This lets us measure how far apart two answers are even when they come from scales of different lengths.

Illustrative Example. Table 1 traces the full pipeline for one OpinionQA question. The question stem is held constant throughout. Three reworded variants illustrate how cardinality, phrasing, and polarity are varied: V1 and V2 are both 4-point bipolar scales with synonymous labels, while V3 extends to five points, adding a neutral midpoint. All three map semantically equivalent top options to $+1.0$, so a consistent model should select $+1.0$ across all variants regardless of the specific wording. The final row shows a shuffled permutation of V1: here, the option previously in position 1 (“Strongly agree”) now occupies position 3. A model relying on positional cues would select option 1 in both cases, yielding $+1.0$ in V1 but -0.33 in S1, and an APD of 1.33 for this single question.

3.2 Model Configurations

We evaluate four open-weight model families, including Llama 3.1 (Grattafiori et al., 2024), Mistral (Jiang et al., 2023), Qwen 2.5 (Qwen et al., 2025), Gemma 2 (Team et al., 2024), each in both a pretrained Base and an Instruction-Tuned (IT) variant. All models range from 7B to 9B parameters, balancing computational feasibility with broad representativeness across pre-training recipes and tokenization strategies. Comparing Base and IT variants within each family isolates the effect of alignment training while holding architecture and scale constant.

All inference uses greedy decoding (`do_sample=False`) to eliminate sampling

Variant	Options (in presentation order)	Normalized values
Question: “How much do you agree that the government should do more to protect the environment?”		
V1 (4-pt, bipolar)	Strongly agree / Somewhat agree / Somewhat disagree / Strongly disagree	+1.0 / +0.33 / -0.33 / -1.0
V2 (4-pt, bipolar)	Fully agree / Partially agree / Partially disagree / Fully disagree	+1.0 / +0.33 / -0.33 / -1.0
V3 (5-pt, bipolar)	Completely agree / Mostly agree / Neither / Mostly disagree / Completely disagree	+1.0 / +0.5 / 0.0 / -0.5 / -1.0
S1 (V1, shuffled)	Somewhat disagree / Strongly disagree / Strongly agree / Somewhat agree	-0.33 / -1.0 / +1.0 / +0.33

Table 1: Variation Engine example. V1 and V2 use different labels for the same 4-point bipolar structure; V3 adds a neutral midpoint. All three map the top option to +1.0. S1 shuffles V1: a primacy-biased model selecting option 1 receives -0.33 instead of +1.0, yielding APD = 1.33 for this question.

noise, ensuring that any observed response variation is attributable to prompt design rather than stochastic generation. Details are shown in Appendix A.

3.3 Prompting Strategies

We evaluate four zero-shot prompt formats as a cross-cutting experimental factor, designed to test robustness to structural label choices that vary across real survey instruments:

- **P1 – Numeric:** Options labeled “1), 2), 3), ...”
- **P2 – Alphabetic:** Options labeled “A), B), C), ...”
- **P3 – Unlabeled:** Options listed as plain text lines, with no labels.
- **P4 – Extended:** Numeric labels augmented with two non-substantive options: “Don’t Know” and “Refused”, mimicking real survey instruments that include refusal codes.

IT models receive all instructions via a single user turn in the model’s chat template, with no separate system prompt (to standardize across model families). Base models receive a raw text-completion prompt ending with a forcing suffix (e.g., “The best answer is option number:”). Full prompt templates are given in Appendix B.

3.4 Consistency Metrics

Rather than measuring accuracy against a ground truth, which is undefined for subjective opinions, we quantify *self-consistency* at the question level. This mirrors the psychometric concept of parallel-forms reliability: a model expressing a stable underlying opinion should produce responses that cluster tightly regardless of which scale formulation is used. For a set of valid normalized responses $V = \{v_1, \dots, v_n\}$ ($n = 5$ for Reworded, $n = 6$ for Shuffled), we compute four metrics:

Average Pairwise Distance (APD). The mean absolute difference across all unique response pairs. APD = 0 indicates perfect consistency; higher values indicate semantic drift across scale variants.

$$\text{APD} = \frac{2}{n(n-1)} \sum_{i < j} |v_i - v_j| \quad (1)$$

Maximum Pairwise Distance (MPD). The worst-case divergence, $\max_{i < j} |v_i - v_j|$. An MPD value of 2.0 represents a complete polarity inversion, i.e., a shift from the most positive response extreme to the most negative for the same question.

Mean Standard Deviation (MSD). The mean of per-question standard deviations across all questions in the dataset. MSD captures average response dispersion and is more sensitive to general instability than to individual extreme cases.

Pearson Correlation (r). Computed between model response vectors across prompt-format pairs (e.g., P1 vs. P2). A high r indicates that the model preserves the relative semantic ranking of questions even when the absolute response format changes, reflecting structural stability of the model’s internal opinion ordering.

Responses failing strict output parsing and non-substantive P4 selections (“Don’t Know”, “Refused”) are excluded. A question is included only if all n variations yielded valid substantive responses, to avoid artificially deflating variance through missing data.

4 Results

4.1 Global Response Patterns: Mechanical Artifacts vs. Semantic Stability

Before computing consistency metrics, we examine the raw response distributions to characterize the systematic biases present in each model class. Figure 2 shows the percentage of responses in each option category for Base models on the Reworded Dataset.

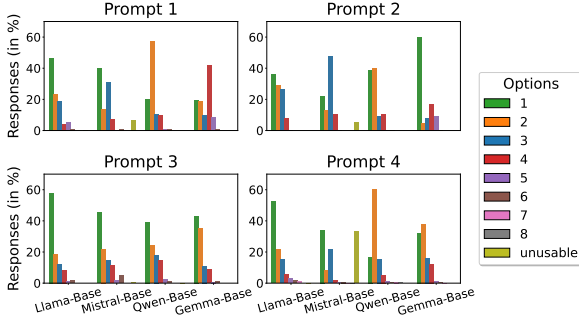


Figure 2: Response category distributions for Base models on the Reworded Dataset. The four subplots within each model panel correspond to the four prompt formats (P1–P4). Valid option indices range from 1 to 6 (up to 8 in P4). Llama-3.1-8B shows consistent primacy bias; Gemma-2-9B shows radical label sensitivity (strong preference for Option 4 in P1, shifts to Option 1 in P2); Mistral-7B-v0.3 accumulates high rates of unusable outputs in P4; Qwen2.5-7B shows moderate and relatively stable preference patterns.

The figure reveals four qualitatively distinct failure modes in Base models, each reflecting a different source of measurement error from a survey methodology perspective:

Primacy bias (Llama-3.1-8B). This model defaults to Option 1 in 35–60% of all responses, depending on the prompt format, regardless of question content. This mirrors the primacy effects documented in human survey research (Krosnick and Alwin, 1987), but in an extreme, mechanically consistent form.

Label sensitivity (Gemma-2-9B). This model strongly favors Option 4 (>40%) under numeric labels (P1), but shifts dramatically to Option 1 (>60%) under alphabetic labels (P2). The preference is not driven by semantic content but by how option labels interact with the model’s learned probability distribution over tokens.

Formatting failures (Mistral-7B-v0.3). This model produces unusable responses at elevated rates in P1 and P2 ($\approx 8\%$) and catastrophically in P4 ($\approx 35\%$), where the unusable category rivals the most selected valid option. This reflects a mismatch between the model’s text-completion training and the structured output format required by survey questions.

Aggregate distributional instability. A reliable survey respondent should express a similar aggregate opinion profile regardless of whether options are presented in their standard order or shuffled. In contrast, Gemma-2-9B Base shows a near-complete inversion of its aggregate opin-

ion distribution between the Reworded (fixed order) and Shuffled conditions: under the Reworded Prompt 1, density is concentrated at the negative extreme (-1.0); under the Shuffled condition, density spikes at the positive extreme ($+1.0$). This polarity flip, driven purely by option reordering, constitutes a fundamental reliability failure under the parallel-forms standard.

In contrast, IT models show near-perfect format adherence (unusable rates $< 0.1\%$), maintain stable response profiles across semantic rewordings, and produce aggregate opinion distributions that overlap substantially between Reworded and Shuffled conditions, demonstrating that alignment successfully grounds outputs in semantic content rather than presentation artifacts.

4.2 Intra-Prompt Consistency: Volatility Analysis

We now quantify how much a model’s response to a given question changes across scale variations within a fixed prompt format.

Average and worst-case divergence. Figure 3 shows APD distributions for all models on the Shuffled Dataset under P1. Base model distributions are markedly diffuse: Mistral-7B-v0.3 spreads density nearly uniformly from APD 0.0 to 1.25 (mean ≈ 0.57), and Gemma-2-9B has a distribution shifted further right, with a median of ≈ 0.63 and a distinct density spike near 0.8. For comparison, an APD of 0.67 corresponds to a model that, on average, moves one full step on a 4-point scale between scale variants. Llama-3.1-8B and Qwen2.5-7B show somewhat better clustering in the 0.2–0.5 range, but retain substantial high-distance tails.

The worst-case metric (MPD) adds further evidence of Base model fragility. Across all Base models on the Reworded Dataset, a non-trivial fraction of questions receive $MPD = 2.0$, representing a complete polarity inversion where the model gives diametrically opposite responses (e.g., “Strongly Agree” and “Strongly Disagree”) to the same question under different scale formulations. For Gemma-2-9B, a bimodal MPD distribution is observed, with density spikes at both 0.0 (perfect stability for some questions) and 2.0 (full inversion for others), indicating that the model’s stability is entirely content-dependent with no general consistency principle.

Comparison to a random-response baseline. To determine whether Base models are merely out-

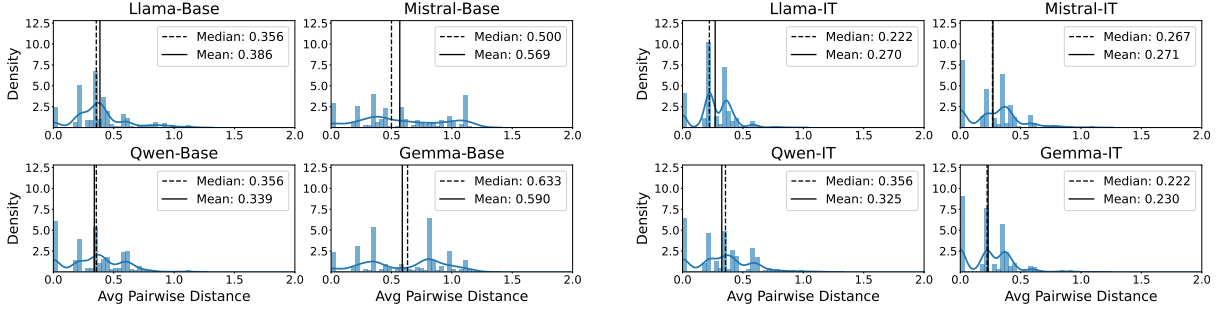


Figure 3: APD distributions on the Shuffled Dataset under P1. **Left:** Base models show highly diffuse distributions, reflecting continuous semantic drift across option permutations. **Right:** Instruction tuning compresses distributions sharply toward 0. The improvement is most dramatic for Gemma-2-9B, which transforms from the most volatile Base model to the most stable IT model.

putting noise, we simulated a uniform-random responder that selects valid scale options with equal probability. On the Reworded Dataset (5 variants), this null baseline yields a mean APD of 0.825, a mean MPD of 1.632, and a MSD of 0.711. On the Shuffled Dataset (6 variants), the mean APD is 0.833, mean MPD is 1.734, and MSD is 0.724. Comparing these null values to the observed Base model metrics (e.g., Mistral’s Shuffled APD of ≈ 0.57 in Figure 3, or the Reworded MSDs of 0.22–0.45 shown in Table 2) confirms that Base models are not merely outputting uniform noise. They retain a meaningful semantic signal on average, albeit a weaker and more volatile one than that of IT models.

The reliability filter effect of instruction tuning. Figure 3 (right) and Table 2 together quantify the stabilization produced by instruction tuning. IT model APD distributions concentrate sharply near 0 across all four families. Table 2 shows that across all models and prompt formats, IT models achieve MSD values roughly half those of their Base counterparts (e.g., Llama-3.1-8B drops from 0.357 to 0.207 under P1; Gemma-2-9B drops from 0.344 to 0.174). Gemma-2-9B-IT achieves the lowest MSD of all models (≈ 0.17 on average), a particularly striking result given that Gemma-2-9B (Base) is among the most volatile, confirming that stability is a consequence of alignment rather than an inherent architectural property.

Prompt format and label effects. An important secondary finding from Table 2 concerns the Unlabeled format (P3). For Base models, removing option labels causes MSD to spike substantially (Mistral-7B-v0.3: 0.453; all others: ≥ 0.41), confirming that labels function as structural anchors

Model	P1	P2	P3	P4
<i>Base Models</i>				
Llama-3.1-8B	0.357	0.315	0.411	0.340
Mistral-7B-v0.3	0.302	0.254	0.453	0.294
Qwen2.5-7B	0.254	0.304	0.415	0.224
Gemma-2-9B	0.344	0.325	0.421	0.229
<i>IT Models</i>				
Llama-3.1-8B-IT	0.207	0.229	0.248	0.203
Mistral-7B-IT	0.221	0.227	0.287	0.206
Qwen2.5-7B-IT	0.232	0.249	0.244	0.241
Gemma-2-9B-IT	0.174	0.172	0.178	0.164

Table 2: Mean Standard Deviation (MSD) on the Reworded Dataset. P1 (Numeric), P2 (Alphabetic), P3 (Unlabeled), P4 (Extended). IT models achieve approximately half the MSD of Base models across all conditions.

that support mapping semantic intent to a discrete output token. Without these anchors, Base models lose the ability to produce consistent selections. IT models are significantly less affected (P3 MSD: 0.178–0.287), reflecting their stronger semantic grounding.

4.3 Removing Labels Can Help When Options Are Shuffled

Comparing the Reworded and Shuffled datasets turns up a pattern that runs against intuition: whether unlabeled options help or hurt consistency depends entirely on whether the option order is fixed or scrambled.

On the Reworded Dataset, where option order is fixed, P3 (Unlabeled) produces the highest MSD of the four formats for both model classes, as discussed above, so dropping labels there clearly hurts. But on the Shuffled Dataset, where option order is randomized, P3 gives the *lowest* MSD of the four formats for every instruction-tuned model

family. As shown in Table 6 (Appendix C.5), Gemma-2-9B-IT achieves $\text{MSD} = 0.121$ under P3 on the Shuffled Dataset, lower than its values for P1 (0.232), P2 (0.187), and P4 (0.220). Similar patterns hold for Llama-3.1-8B-IT (P3: 0.146 vs. P1: 0.277) and Qwen2.5-7B-IT (P3: 0.206 vs. P1: 0.310).

Here is how we read this. Under labeled prompts (P1, P2) with shuffled options, instruction-tuned models are pulled in two directions at once: a learned habit of favoring certain labels (say, a tendency to answer “A” or “1” regardless of content) and a preference for whichever option actually matches their opinion. When shuffling happens to put the model’s preferred content behind an unfavored label, these two pulls disagree, and the resulting answers become less stable. Take the labels away, and that conflict simply disappears: the model has nothing left to lean on but the content itself, and the content is the same no matter how the options are ordered, so consistency goes up.

This lines up with the Multiple Choice Symbol Binding account of Xue et al. (2024): the failure sits at the connection between a label and its content, so removing the label sidesteps the failure for models that understand the content well enough to rely on it. Base models do not show this effect, because they never developed strong enough content understanding to make use of the missing labels, so P3 stays their worst format either way.

4.4 Inter-Prompt Structural Robustness

We now assess whether models preserve a consistent logical stance when the structural format of the prompt itself changes, independent of option order and wording variations.

Categorical consistency. Figure 4 shows the distribution of categorical agreement levels across all four prompt formats on the Reworded Dataset. Perfect agreement (“4 Same”) means the model selects the same response category for all four prompt formats for a given question.

Base models exhibit clear logical fragmentation. Gemma-2-9B is the most fragile: “2 Same” is its dominant category ($\approx 45\%$), and it records the highest rate of total disagreement (“4 Different”: $\approx 5\%$). Mistral-7B-v0.3 achieves perfect categorical agreement in fewer than 20% of questions. Llama-3.1-8B and Qwen2.5-7B perform somewhat better, with “4 Same” around 32%, but “3 Same” or fewer is still the majority outcome.

Instruction tuning induces a decisive shift toward structural stability. “4 Same” becomes the dominant outcome for all four IT families, ranging from 43% (Mistral-7B-IT) to 62% (Llama-3.1-8B-IT). The “4 Different” category drops to $<1\%$ for all IT models. The residual “3 Same” category ($\approx 27\text{--}34\%$) primarily reflects structurally valid behavior: when a model appropriately selects “Don’t Know” or “Refused” in P4, this prevents a four-way match even though the model’s substantive response in P1–P3 is perfectly consistent.

Correlational stability. Table 3 presents Pearson’s r between response vectors across all prompt-format pairs on the Shuffled Dataset, measuring whether the model’s relative opinion ranking across questions is preserved under format changes.

Model	1–2	1–3	1–4	2–3	2–4	3–4
<i>Base Models</i>						
Llama-3.1-8B	0.545	0.354	0.594	0.441	0.511	0.413
Mistral-7B-v0.3	0.309	0.183	0.496	0.191	0.269	0.244
Qwen2.5-7B	0.608	0.428	0.849	0.431	0.572	0.400
Gemma-2-9B	0.468	0.155	0.499	0.312	0.557	0.378
<i>IT Models</i>						
Llama-3.1-8B-IT	0.797	0.636	0.796	0.686	0.778	0.644
Mistral-7B-IT	0.730	0.709	0.901	0.627	0.728	0.718
Qwen2.5-7B-IT	0.886	0.783	0.921	0.777	0.882	0.780
Gemma-2-9B-IT	0.849	0.733	0.856	0.784	0.824	0.717

Table 3: Pearson’s r between prompt-format pairs on the Shuffled Dataset. IT models maintain high logical ranking stability ($r \approx 0.70\text{--}0.92$). Base models suffer severe degradation, especially in pairs involving P3. The P1–P4 pair is consistently the most stable, reflecting structural similarity.

IT models maintain high structural stability ($r \approx 0.70\text{--}0.92$) regardless of format pair. Base models’ response vectors are substantially scrambled when format changes ($r \approx 0.15\text{--}0.85$). Two consistent structural patterns emerge. First, P1–P4 yields the highest correlations across all models: for IT models, r ranges from 0.796 to 0.921, and even for Base models Qwen2.5-7B achieves 0.849. This confirms that models recognize the structural similarity between numeric and extended-numeric formats and maintain opinions when refusal options are added. Second, prompt pairs involving P3 consistently yield the lowest correlations, often 0.1–0.2 points below labeled pairs, confirming that unlabeled prompts require a qualitatively different processing pathway.

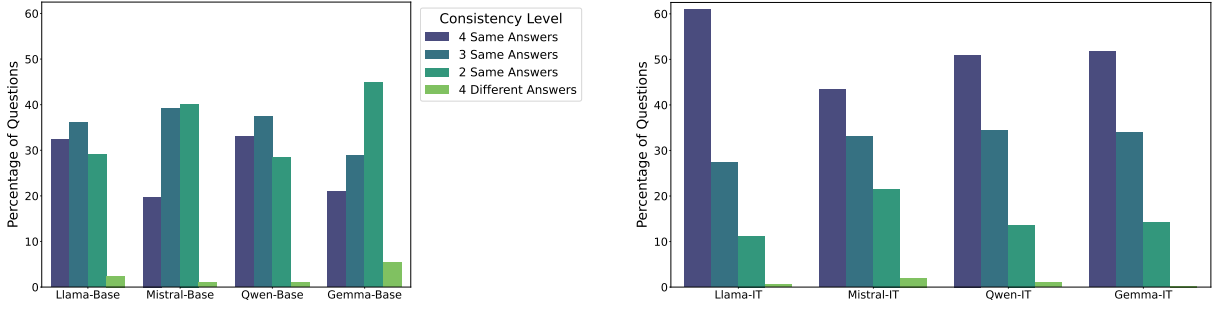


Figure 4: Categorical consistency across four prompt formats on the Reworded Dataset. **Left:** Base models exhibit logical fragmentation, with perfect agreement rarely dominant. **Right:** Instruction tuning shifts the dominant outcome to perfect agreement across all four families. The “3 Same” residual in IT models largely reflects structurally valid selections of “Don’t Know” or “Refused” in P4.

5 Discussion

5.1 Psychometric Interpretation: Reliability as an Engineered Property

From the perspective of Classical Test Theory, our results can be summarized concisely: Base models fail to meet the parallel-forms reliability standard, while IT models approach it but do not fully achieve it. The measurement error introduced by scale design is substantial for Base models (MSD 0.22–0.55), meaning that a large fraction of the observed variation in Base model survey responses reflects properties of the measurement instrument rather than the model’s underlying stance. IT models reduce this error substantially, but residual scale sensitivity (MSD 0.16–0.30) means that even aligned models do not yet meet the reliability standards typically required for survey instruments in social science research.

Importantly, this distinction is not primarily a matter of model architecture or parameter count. Within each of the four families, the IT variant dramatically outperforms the Base variant on every consistency metric. The effect of alignment training (SFT + RLHF) is a fundamental behavioral shift: Base models optimize for next-token prediction and are therefore sensitive to surface-level statistical regularities in their pre-training data, such as the frequency of particular option labels or the distribution of answers appearing at particular positions (Zhao et al., 2021). IT models are additionally trained to follow semantic instructions, which reduces sensitivity to these surface features and grounds outputs more directly in the semantic content of the question (Ouyang et al., 2022).

While our evaluation is explicitly scoped to models in the 7B-9B parameter class, prior literature suggests that simply scaling up parameter counts

does not automatically resolve format sensitivity. Elazar et al. (2021) demonstrate that while larger models consistently achieve higher accuracy, semantic inconsistencies under paraphrasing remain a pervasive issue regardless of scale. Similarly, Rupprecht et al. (2025) note that while higher-capacity models exhibit marginally greater stability on survey items, no model size proves inherently robust to scale or phrasing perturbations.

Finally, while we lack human baselines for these specific reworded and shuffled variants, typical human parallel-forms reliability for well-validated Likert scales generally produces Pearson correlations in the range of $r \approx 0.70$ to 0.85 (Lord et al., 1968; Nunnally, 1978). By this benchmark, our Instruction-Tuned models ($r \approx 0.70$ - 0.92) operate at or slightly above the threshold of typical human reliability, whereas Base models ($r \approx 0.30$ - 0.60) fail it completely. This contextualizes the performance of IT models not just as a mechanical improvement, but as achieving a baseline level of consistency comparable to human survey respondents.

5.2 Symbol Binding, Labels, and Prompt Design

The pattern from Section 4.3, where removing labels helps once the options are shuffled, points to a subtle but practically important difference in how instruction-tuned models handle labeled versus unlabeled prompts. Labels do two different jobs depending on the model. For base models, they are essential anchors that connect an abstract output token to a specific answer option; take them away, and that connection breaks down. For instruction-tuned models, though, labels introduce a second signal that can pull against the model’s actual con-

tent preference, especially once the options have been shuffled. This tension between choosing by label and choosing by content fits with the symbol-binding failures documented by [Xue et al. \(2024\)](#), but here the fix turns out to be removing the label entirely rather than improving how it is bound to content.

There is likely also an alignment-training angle here. RLHF may leave models with a mild habit of favoring certain label tokens (“A” or “1”) as a kind of learned response style, separate from what the content actually says. When the options are shuffled, that habit works against the model’s real semantic preference and shows up as inconsistency. Drop the labels, and the habit has nothing to attach to. Some of the stability we see in instruction-tuned models is probably genuine semantic reasoning, and some of it may be a form of learned compliance ([Sharma et al., 2024](#)), but either way, our results show that the net effect of alignment training on survey reliability is clearly positive across the board.

6 Conclusion

We set out to test whether LLMs meet the parallel-forms reliability standard expected of survey instruments, by systematically varying Likert-scale wording, cardinality, polarity, and option order. Our main finding is straightforward: reliability in LLM-based survey research is not something 7-9B models come with out of the box, it is produced by instruction tuning and remains sensitive to how prompt labels are formatted. Base models fail basic reliability checks and should not be used as synthetic survey respondents. Instruction-tuned models cut response variance by roughly half and show meaningful structural robustness, but some scale sensitivity remains even for them. We also find a less obvious interaction between prompt format and option shuffling, where removing labels can improve reliability once option order is randomized, which has direct implications for how synthetic surveys should be designed. Future work should extend this evaluation to paraphrasing the question stems themselves, to larger models, to non-English survey corpora, and, most importantly, should compare LLM reliability against human parallel-forms reliability coefficients to see whether current instruction-tuned models meet the standards we apply to human respondents.

Limitations

Our evaluation is subject to several limitations. First, we rely on strict output parsing, which may conflate formatting failures with genuine reasoning failures ([Molfese et al., 2025](#)). Penalizing a model for selecting “Don’t Know” in P4, for instance, ignores that expressing uncertainty can reflect appropriate calibration rather than inconsistency ([Kadavath et al., 2022](#)). Second, Mistral-7B-v0.3’s high unusable rate in P4 means that the consistency metrics for that model are computed over a filtered, potentially non-representative subset of questions. Third, all inference uses greedy decoding; robustness under stochastic sampling and higher temperatures remains unexplored. Fourth, the study is limited to English-language questions and four 7-9B parameter models; consistency patterns may differ for larger models or non-English corpora. Fifth, and most critically, we lack human baselines answering the same reworded and shuffled variants, making it impossible to determine whether current LLM consistency levels are subhuman, comparable to human reliability, or in some cases superior (“hyper-accurate”) but brittle in qualitatively different ways ([Aher et al., 2023](#)). Future work should also extend perturbations to question-stem paraphrasing ([Elazar et al., 2021](#)) and to a broader range of scale cardinalities and polarities ([Menold, 2021](#)).

Ethical Considerations

We use the English OpinionQA dataset ([Santurkar et al., 2023](#)), which is freely available for research. Our extended dataset is released exclusively for non-commercial research purposes and respects all intellectual property and privacy requirements of the original dataset. Experimental compute was performed on institutional GPU infrastructure. No personal information was collected. The authors acknowledge the use of ChatGPT solely for grammar correction and text coherence, and for assistance with coding.

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A Implementation Details

A.1 Data Preparation and Filtering

Starting from the 1,498 OpinionQA questions, a two-stage manual review reduced the dataset to 1,235 items suitable for scale-variation experiments. In the first stage, questions were excluded if they implicitly assumed a specific respondent background (e.g., “While growing up, how often did you . . .”; “How much . . . would you say is in your neighborhood”), because such items would require the model to adopt a demographic persona rather than express a generalized opinion. In the second stage, any remaining question whose options were structured as substantive statements rather than scalar degrees (e.g., “Which of the following statements comes closer to your view?” with two distinct policy positions) was reformulated into a Likert-administrable question. This involved converting the original stem into a statement and prefacing it with “How much do you agree or disagree with the following statement?”, then treating each original option as a separate item. This adaptation preserved the substantive content of the original survey while enforcing a uniform Likert-scale structure.

Content dimension classification was performed with an automated Python script combining keyword matching and Regular Expressions. Question stems were matched against 15 dimension-specific keyword lists (e.g., stems containing *agree/disagree/support/oppose* were assigned to the “Agreement” dimension; those containing *how likely* or *probability* to “Likelihood”; those with comparative terms like *better/worse* to “Better/Worse”). The resulting distribution is shown in Table 4. Each question type was then paired with a set of five semantically equivalent but differently worded answer option sets. The Agree-Disagree (AD) format was adopted as the primary scale structure because it allows a uniform scale design across diverse question topics, isolating the effect of scale parameters (cardinality, phrasing, polarity) from question semantics, in line with the goal of studying instrument effects rather than item content effects. We acknowledge that AD scales impose an additional cognitive mapping step compared to item-specific scales (Dykema et al., 2022), but this limitation is secondary to the experimental goal of systematic scale variation.

For the Shuffled Dataset, five random permutations of the original option order were generated

for each question using fixed random seeds (2, 7, 10, 67, 101), yielding six orderings per item (original plus five shuffled). The Shuffled Dataset uses scales of up to five options, as the factorial growth of 6-option permutations ($6! = 720$) makes systematic sampling impractical within the experimental budget.

A.2 Hardware and Decoding

All experiments were executed on a single NVIDIA A100 (80 GB) GPU, using the Hugging Face transformers library (Wolf et al., 2020). Models were loaded in native precision (`dtype="auto"`, typically `bfloat16` or `float16`) to avoid performance degradation from quantization. Greedy decoding (`do_sample=False`) was used throughout, with `max_new_tokens=50`. Greedy decoding ensures that any observed variation across items is attributable solely to the changes in scale wording or option order, not to stochastic sampling.

A.3 Inference Pipelines

IT models received prompts via the model’s native chat template in a single user turn (no system prompt). Omitting the system prompt was a deliberate choice: adding a persona-shaping system message (e.g., “You are a helpful assistant”) would introduce an uncontrolled variable across model families, which implement system-prompt parsing differently. Base models received a raw text-completion prompt ending with a forcing suffix (e.g., “The best answer is option number:”). Without this suffix, Base models tend to continue the list of options as if completing a document, rather than selecting one. Model-specific end-of-sequence tokens were added to the stopping criteria to prevent the model from generating additional text after the answer token.

A.4 Content Dimensions

Table 4 shows the distribution of questions across the 15 content dimensions after filtering and adaptation. The “Agreement” dimension is the most prevalent (32.7%), reflecting the design of the original Pew Research surveys, which frequently ask respondents to express their degree of agreement with policy positions or factual statements. The next-largest dimension is “Quantity” (17.3%), capturing items about the appropriate amount or level of something (e.g., government spending, personal frequency of behavior). The remaining 13 dimensions each account for under 9% of the dataset,

providing a diverse coverage of opinion-related response formats while keeping the classification scheme manageable.

Content Dimension	Count	%
Agreement	404	32.7
Quantity	214	17.3
Better / Worse	101	8.2
Importance	91	7.4
Likelihood	85	6.9
Positive / Negative	67	5.4
Good / Bad	50	4.0
Magnitude of Problem	48	3.9
Frequency	38	3.1
Performance (“How Well”)	38	3.1
Priority	35	2.8
Acceptance	31	2.5
Reason	15	1.2
Concern	11	0.9
Increase / Decrease	7	0.6
Total	1,235	100.0

Table 4: Content dimension distribution ($N = 1,235$).

B Prompt Templates

Design Rationale. The prompting strategy pursues two goals simultaneously: reliably extracting a single answer token, and minimizing extraneous biases not under study. The opening framing “You are taking a survey. There are no right or wrong answers” serves a dual purpose. It signals that the task is opinion-based rather than factual, reducing the likelihood of content-moderation refusals on politically sensitive items, and it suppresses the tendency of instruction-tuned models to hedge or justify rather than commit to a scalar response.

A **zero-shot** approach was adopted throughout. Including few-shot demonstration examples would introduce an additional positional confound: the label positions in the example items could prime the model toward particular options (e.g., a demonstration always selecting option A could bias subsequent selections). Zero-shot evaluation ensures that any systematic option preference reflects the model’s own prior rather than a learned heuristic from the context.

For **IT models**, all content was delivered in a single user turn without a system prompt, as described in Appendix A. The output format constraint (“Provide exactly one [label] only”) was added after pilot experiments revealed that instruction-tuned models frequently returned verbose explanations or cited multiple options unless explicitly instructed to output a single token.

For **Base models**, the instruction block was removed and replaced by a forcing suffix (“The best answer is option [label]:”) appended directly after the option list. This text-completion framing exploits the base model’s training objective: it predicts the next token following the suffix, which is typically the answer label.

The four prompt format variants (P1–P4) differ only in how options are labeled and presented. **P1 (Numeric)** uses “(1) ... (2) ...” enumeration; **P2 (Alphabetic)** uses “(A) ... (B) ...”; **P3 (Unlabeled)** lists options as plain newline-separated text without any prefix; **P4 (Extended)** augments P1 with two additional options (“Don’t Know” / “Refused”) labeled $N+1$ and $N+2$, mapped to exclusion codes during analysis. For P3, the output-format instruction for IT models is abbreviated to “Please answer with ONLY one of the answer options given above” without specifying a label type, since no labels exist. Base models receive no forcing suffix under P3. The detailed template is shown in Figure 5.

C Supplementary Results

C.1 Dataset Scale Characteristics

Table 5 shows the distribution of scale types across the two experimental datasets. The Reworded Dataset includes 6-point scales (bipolar and unipolar) that do not appear in the Shuffled Dataset. This asymmetry arises because 6-point scales generate 720 possible permutations ($6!$), making systematic random sampling computationally and analytically impractical. The five seeds used for the Shuffled Dataset are instead applied to scales of up to five options, which yield 120 permutations at most. 4-point scales dominate both datasets (58.6% of Reworded, 78.7% of Shuffled) because the majority of content dimensions are served by a standard 4-point bipolar or unipolar structure. The 5-point bipolar format accounts for the next largest share, capturing dimensions (such as Importance or Likelihood) for which a neutral midpoint is semantically appropriate. The co-occurrence of variable cardinalities in the Reworded Dataset is a key driver of the smoother APD distributions described in Appendix C.7: when a 4-point and a 5-point variant are compared, the normalized distance values can take on more intermediate steps than when comparing two variants of identical length.

IT Models (Chat Template)	Base Models (Text Completion)
You are taking a survey. There are no right or wrong answers. Please choose the one option that best fits your view. The order of the options does not imply any ranking. Output format: Provide exactly one [label] only (no words, no punctuation, no quotes). Question: [question] Options: [options] Please answer with ONLY one of the answer options given above, and with only the [label] corresponding to the option you choose.	Question: [question] Options: [options] The best answer is option [label]:

Figure 5: Core prompt templates. [label] is substituted with “number” (P1), “letter” (P2), or “option” (P3). For P3 (Unlabeled), the IT prompt omits the output-format line and truncates at “...given above.”; no forcing suffix is used for Base models under P3. For P4, the option block is extended with numbered “Don’t Know” and “Refused” entries.

Scale Type	Reworded		Shuffled	
	N	%	N	%
4-pt Unipolar	1,858	30.1	3,408	46.0
4-pt Bipolar	1,760	28.5	2,424	32.7
5-pt Bipolar	1,375	22.3	1,350	18.2
5-pt Unipolar	635	10.3	228	3.1
6-pt Bipolar	512	8.3	—	—
6-pt Unipolar	35	0.6	—	—

Table 5: Scale type distributions across the two datasets. P4 extends all scales by +2 options (“Don’t Know” / “Refused”), not reflected in these counts.

C.2 IT Model Response Distributions

Figure 6 shows the response category distributions for IT models on the Reworded Dataset, and should be read alongside Figure 2 in the main paper, which covers Base models. The contrast is stark along two dimensions.

First, the **unusable rate** for IT models is negligible ($< 0.1\%$) across all four prompt formats, whereas Base models exhibit unusable rates ranging from around 8% under standard formats to nearly 35% under P4 for Mistral. This confirms that instruction tuning reliably enforces output format constraints: IT models correctly interpret the label-selection instruction and respond with a single token, even under the more complex P3 (Unlabeled) and P4 (Extended) formats that confuse base models.

Second, IT models display markedly **stable semantic profiles** across P1–P4. Unlike Base mod-

els, which shift their modal preference dramatically depending on whether options are labeled numerically or alphabetically (e.g., Gemma-2-9B selects Option 4 most often under P1 but shifts to a sharp spike at Option 1 under P2), IT models maintain broadly consistent response distributions regardless of the label type used. This structural stability is the behavioral correlate of the lower MSD values reported in Table 2 of the main paper, and suggests that instruction tuning reduces sensitivity to surface-level label format rather than merely suppressing invalid outputs.

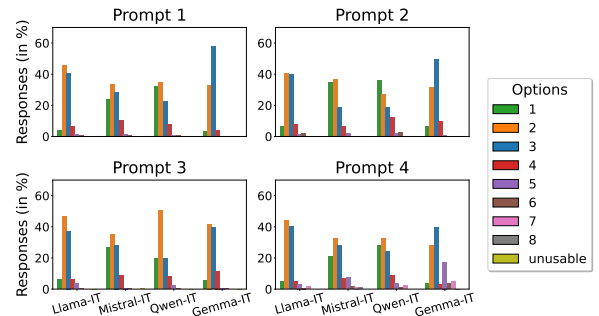


Figure 6: Response category distributions for IT models on the Reworded Dataset. Unusable rates are negligible ($< 0.1\%$) and response profiles are consistent across all four prompt formats, in sharp contrast to the Base model distributions in Figure 2.

C.3 Model-Specific Behavior Profiles

Beyond the general Base/IT contrast, our results reveal instructive model-specific patterns that illuminate sources of instability. **Gemma-2-9B (Base)** represents the most dramatic case of format-driven instability. Under Prompt 1 (Numeric), responses concentrate at the negative extreme (-1.0) on the Reworded Dataset, but flip to the positive extreme ($+1.0$) when shuffled. This polarity inversion suggests a strong position bias when numerical labels no longer reliably identify options. After instruction tuning, Gemma-2-9B-IT becomes the most consistent model, confirming that alignment training effectively resolves this sensitivity.

Qwen2.5-7B (Base) exhibits atypically high intrinsic stability, with inter-prompt correlations ($r = 0.849$) approaching IT-level performance. Its aggregate distributions remain similar between Reworded and Shuffled conditions, likely reflecting high-density structured MCQA reasoning in pre-training. **Mistral-7B-v0.3 (Base)** is the least structurally consistent Base model, with inter-prompt correlations frequently below 0.25. Its IT variant shows improvement but remains the least consistent IT model, suggesting alignment training is less effective at resolving underlying symbol-binding tensions.

C.4 The Qwen2.5 Base Model Anomaly

Qwen2.5-7B (Base) presents an atypical pattern among Base models. While other Base models show large distributional divergences between Reworded and Shuffled conditions, Qwen2.5-7B maintains relatively overlapping distributions across both datasets. Its inter-prompt correlations on the Reworded Dataset reach P1–P4: $r = 0.849$ and P1–P3: $r = 0.575$, values that are comparable to those of some IT models from other families (cf. Table 7) and substantially higher than the other Base models (e.g., Mistral Base P1–P3: $r = 0.408$, Gemma Base P1–P3: $r = 0.374$). Its MSD on the Reworded Dataset (P1: 0.254, lowest among all Base models) and the Shuffled Dataset (P1: 0.323, again lowest) further confirm this outlier status.

We attribute this stability to a high density of structured MCQA content in Qwen’s pre-training data. Alibaba Cloud’s Qwen series is trained on a corpus with a reportedly large fraction of question-answering and instruction-style data (Qwen et al., 2025), which may pre-train a rudimentary form of symbol binding even without explicit align-

ment. Figures 7 and 8 illustrate this: while other Base models show the blue (Reworded) and orange (Shuffled) projected-response distributions drifting substantially apart (indicating that option shuffling changes the apparent semantic opinion), Qwen’s distributions overlap substantially across all four prompt formats.

The IT variant of Qwen further smooths and narrows the distributions but provides a smaller absolute improvement than seen in other model families. For instance, Llama’s MSD drops by roughly 42% from Base to IT on P1 ($0.357 \rightarrow 0.207$), whereas Qwen’s improvement is only around 19% ($0.254 \rightarrow 0.232$), suggesting diminishing marginal returns of alignment for a model that already exhibits partial stability at the pre-training stage.

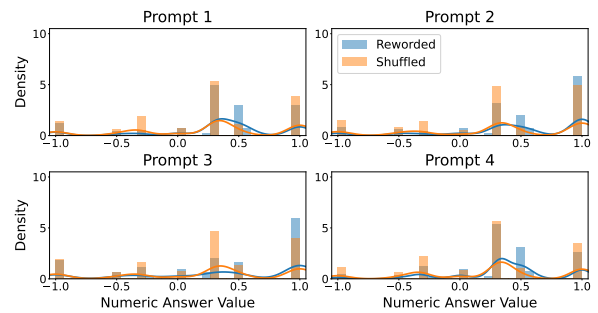


Figure 7: Projected response distributions for Qwen2.5-7B (Base). Blue: Reworded Dataset; Orange: Shuffled Dataset. The two distributions overlap substantially across all four prompt formats, unlike most Base models, which diverge sharply when options are shuffled.

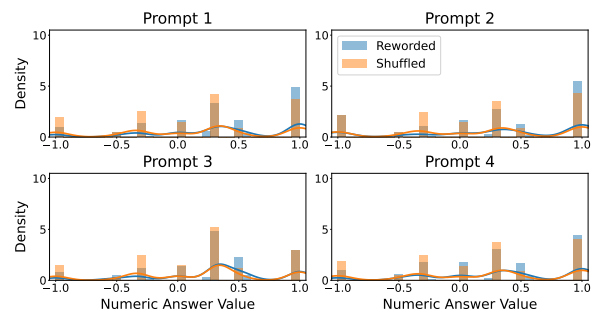


Figure 8: Projected response distributions for Qwen2.5-7B-Instruct (IT). Alignment further narrows the distributional spread, but the high baseline overlap observed in the Base model means the marginal gain from instruction tuning is smaller than for other families.

C.5 MSD on the Shuffled Dataset

Table 6 reports the MSD on the Shuffled Dataset and provides the complete quantitative basis for the Label-Free Consistency Paradox discussed in

Section 4.3 of the main paper.

For IT models, the P3 (Unlabeled) column is bolded because it achieves the lowest MSD for every model family. The reductions are substantial: Gemma-2-9B-IT drops from 0.232 (P1) to 0.121 (P3), a reduction of 47.8%; Llama-3.1-8B-IT drops from 0.277 (P1) to 0.146 (P3), a reduction of 47.3%; Qwen2.5-7B-IT drops from 0.310 (P1) to 0.206 (P3), a reduction of 33.5%. Even Mistral, which shows elevated P3 variance in the Reworded Dataset (Table 2), achieves its second-lowest MSD under P3 in the Shuffled condition (0.240 vs. 0.263 for P1).

The mechanism is as follows. Under labeled formats (P1, P2, P4), shuffling the options creates a conflict between the model’s positional prior (e.g., a tendency to select option “A” or “1”) and its semantic preference (e.g., a tendency to select content meaning “Agree”). Because these two cues point to different options after shuffling, the model’s outputs become inconsistent across orderings. Under P3, no labels are present, so the positional prior is over raw text position rather than a symbol. Since IT models are better calibrated to attend to content, the removal of labels forces them to rely more directly on semantic information, which is invariant across orderings. The result is higher consistency under the very condition that most disrupts labeled formats.

For Base models, this paradox does not materialize systematically. Base model P3 values show no consistent pattern: Llama P3 (0.445) is the highest across its prompts, whereas Mistral P3 (0.365) is the lowest. This is expected: Base models lack reliable symbol binding in the first place, so removing labels does not alleviate a label-induced conflict but instead eliminates the only structural cue that anchored their responses.

C.6 Full Correlation Matrix (Reworded Dataset)

Table 7 reports all six pairwise Pearson correlations between prompt formats on the Reworded Dataset. Several structural patterns are noteworthy.

First, **P1–P4 correlations are consistently the highest** across all model families, for both Base and IT variants (e.g., Mistral-IT: 0.913; Llama-IT: 0.894; Gemma-IT: 0.897). This reflects the structural similarity between these two formats: P4 is a direct extension of P1 (numeric labels) with two additional options appended, so models that have converged on a numeric-label strategy under P1

Model	P1	P2	P3	P4
<i>Base Models</i>				
Llama-3.1-8B	0.367	0.403	0.445	0.328
Mistral-7B-v0.3	0.503	0.400	0.365	0.482
Qwen2.5-7B	0.323	0.361	0.363	0.330
Gemma-2-9B	0.553	0.346	0.415	0.378
<i>IT Models</i>				
Llama-3.1-8B-IT	0.277	0.233	0.146	0.274
Mistral-7B-IT	0.263	0.320	0.240	0.249
Qwen2.5-7B-IT	0.310	0.277	0.206	0.273
Gemma-2-9B-IT	0.232	0.187	0.121	0.220

Table 6: MSD on the Shuffled Dataset. Bold P3 values for IT models confirm the Label-Free Consistency Paradox: removing option labels yields the lowest inter-ordering variance for all four IT model families, even though P3 is the most volatile format in the Reworded Dataset.

apply essentially the same logic under P4, differing only in whether they ever select the “Don’t Know” or “Refused” options.

Second, **pairs involving P3 (Unlabeled) tend to produce lower correlations**, particularly P1–P3 and P3–P4. For IT models, P1–P3 correlations range from 0.701 (Llama) to 0.778 (Gemma), compared to P1–P4 values of 0.894–0.897. This reflects the fundamentally different structural demands of P3, which requires the model to parse and select from plain-text options rather than binding a numeric or alphabetic symbol to content. The label-free format creates a distributional shift in response tendencies (visible in the IT MSD data of Appendix C.5), which lowers correlation with labeled formats.

Third, all correlations in Table 7 are substantially higher than their counterparts in the Shuffled correlation matrix (Table 3 in the main paper). This confirms at the level of individual-question response profiles that semantic rewording is a milder perturbation than positional shuffling: models preserve the relative direction of their opinions across reworded variants better than across shuffled orderings. For IT models, many Reworded correlations exceed 0.80, whereas Shuffled correlations rarely approach this range.

C.7 APD Distributions: Reworded Dataset

Figures 9 and 10 show APD distributions on the Reworded Dataset under P1, and complement the Shuffled Dataset distributions in the main paper.

The most salient structural difference between the two datasets is the shape of the distributions. The **Shuffled Dataset produces discrete spikes**

Model	1-2	1-3	1-4	2-3	2-4	3-4
<i>Base</i>						
Llama-3.1	0.580	0.398	0.644	0.464	0.536	0.461
Mistral-7B	0.485	0.408	0.454	0.354	0.336	0.278
Qwen2.5-7B	0.608	0.575	0.750	0.467	0.560	0.481
Gemma-2-9B	0.534	0.374	0.564	0.473	0.627	0.504
<i>IT</i>						
Llama-3.1	0.820	0.701	0.894	0.712	0.797	0.683
Mistral-7B	0.707	0.722	0.913	0.646	0.709	0.703
Qwen2.5-7B	0.818	0.743	0.883	0.737	0.799	0.726
Gemma-2-9B	0.860	0.778	0.897	0.820	0.841	0.749

Table 7: Pearson’s r across all prompt-format pairs on the Reworded Dataset. All values exceed their Shuffled counterparts (Table 3), confirming that semantic rewording is less disruptive than positional permutation. P1–P4 pairs are consistently the highest due to structural similarity between numeric-label and extended-numeric formats; pairs involving P3 (Unlabeled) are consistently lower.

at a small set of APD values. This is a direct mathematical consequence of fixed scale cardinality: when all five or six variants of a question share the same scale length, the normalized values at which responses can land are constrained to a finite set (e.g., for a 4-point scale mapped to $\{-1.0, -0.33, 0.33, 1.0\}$, the only possible pairwise distances are 0, 0.67, 1.33, and 2.0). As a result, the APD for each question takes on one of a small number of discrete values, creating visible spikes in the density plot at those locations.

The **Reworded Dataset produces smoother, more continuous distributions**. Because the five reworded variants for a given question may draw from scales of different cardinalities (e.g., a 4-point variant paired with a 5-point variant), the set of possible normalized distances is richer. Cross-cardinality pairings generate intermediate values that do not appear in any single-cardinality comparison, effectively densifying the support of the APD distribution and smoothing out the discrete spikes.

For Base models (Figure 9), the distributions are wide and exhibit substantial density in the high-APD region, with a non-trivial fraction of questions producing APDs above 1.0, indicating near-maximal scale traversals triggered by mere wording changes. For IT models (Figure 10), the distributions are strongly compressed toward 0, with most questions achieving APD values below 0.5. The smooth shape is preserved because the variable scale cardinalities apply equally to Base and IT models; the compression reflects reduced sensitivity to wording rather than any change in scale

structure.

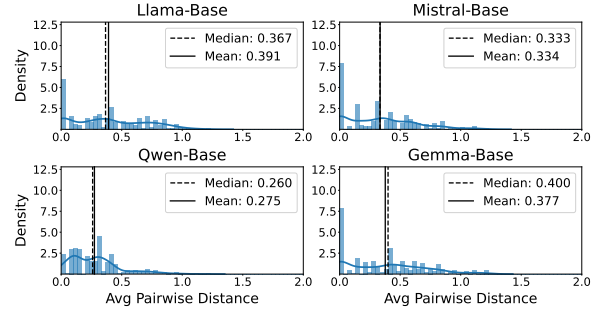


Figure 9: APD distributions for Base models on the Reworded Dataset (P1). The smooth, continuous shape arises from variable scale cardinalities across reworded variants. Wide dispersion and substantial density above APD = 1.0 indicate that wording changes alone can induce near-complete polarity reversals in Base models.

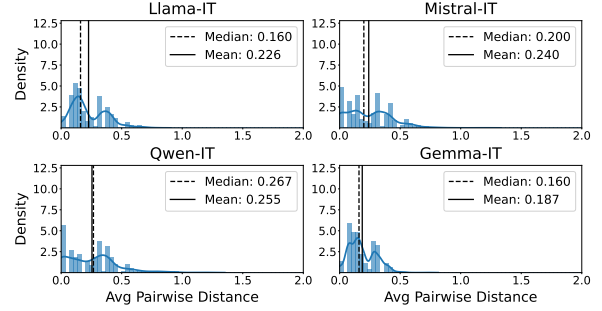


Figure 10: APD distributions for IT models on the Reworded Dataset (P1). The smooth curve shape is maintained by the same variable-cardinality structure, but distributions are strongly compressed toward 0 relative to Base models, reflecting substantially reduced sensitivity to semantic rewording.